

2.2 Data Description

This study used long-term environmental monitoring data obtained from two primary sources: the Oil Sands Monitoring (OSMP) data portal and the Water Quality Portal (WQP). The OSMP portal provides environmental monitoring data across the Athabasca oil sands region, including surface water quality measurements collected as part of ongoing monitoring initiatives. These datasets include chemical parameters relevant to assessing the potential impact of industrial activity on freshwater systems. Additional data were obtained from the Water Quality Portal, which compiles water quality measurements from various monitoring programs and laboratories across aquatic systems. While the sources include both chemical and physical water parameters, differences in data structure, reporting formats, and measurement approaches were observed between the two portals.

Using multiple data sources allowed for broader spatial coverage across lakes, but it also introduced variability in sampling methods and data consistency (Gray et al., 2018). As a result, additional preprocessing steps were required to standardize units, align variable names, and ensure comparability across datasets. Differences between “total” and “total recoverable” measurements required careful consideration. Total concentrations represent the full amount of a substance present in a sample, including both dissolved and particulate-bound forms (Ademollo et al., 2012), while total recoverable concentrations are based on partial digestion methods and may not fully capture substances tightly bound to particulates (Bowie et al., 2010). This distinction means that total recoverable values can underestimate true concentrations but may better reflect what is environmentally available under natural conditions (Harmsen, 2007). Accounting for these differences was necessary to ensure that comparisons between lakes were meaningful and not driven by inconsistencies in measurement methods.

This study focuses specifically on surface water chemistry data, as consistent sediment records were not available across all lakes. While surface water data provides insight into current environmental conditions, it is more sensitive to short-term variability and may not capture long-term contamination trends as effectively as sediment-based analyses (Collins et al., 2004).

The compiled dataset is structured in a wide format, where each row represents a single sampling event and columns correspond to measured chemical parameters, site metadata, and temporal information. Key metadata fields included the following: Sampling Timestamp, Site Name, Lake, and Distance Group, with chemical variables recorded in their own individual columns. The combined data collection spans approximately from 1972 to 2025, although temporal coverage varies across the lakes due to inconsistent sampling frequencies and individual monitoring programs. The obtained data consists of surface water sampling which include chemical and nutrient parameters relevant to the water quality assessment. The original datasets were obtained in a tabular format (CSV), where the individual files contained measurements for specific parameters of the lake. These datasets were combined into a unified

structure for further analysis. The final dataset was structured such that each row represents a unique sampling event defined by date, location/distance group while columns represent chemical and water quality variables. A summary of the key characteristics of the compiled dataset is provided in Table 2.2.1.

A wide range of water chemistry parameters were included in the analysis dependant on the data availability across monitoring programs. These parameters capture key aspects of lake water quality, including major ions, nutrients, metals, and general environmental indicators. The variables included in this analysis are summarized in Table 2.2.2 and grouped into metals, nutrients, major ions, and physical/chemical indicators based on their environmental function. These parameters were selected based on their availability in the compiled datasets and their relevance to assessing water quality and potential environmental impacts. However, not all variables were consistently measured across all lakes or time periods, resulting in uneven data coverage throughout the dataset.

Several data quality challenges were identified due to the integration of multiple data portals. Monitoring programs such as the Oil Sands Monitoring (OSMP) portal and the Water Quality Portal (WQP) introduced inconsistencies in sampling methodologies, laboratory procedures, and parameter availability across lakes. Individual datasets contained varying subsets of chemical and nutrient variables, creating challenges when integrating data across sources. Additionally, parameter naming conventions differed between datasets, with similar measurements recorded under different variable names. For example, turbidity was reported using multiple units (e.g., turbidity_JTU and turbidity_NTU), and nitrogen measurements were recorded in various forms (e.g., nitrite, nitrate, and NO_3/NO_2). These inconsistencies required standardization and merging of equivalent variables to improve consistency and reduce missing values. Despite these efforts, the merged dataset retained a substantial number of missing values due to inconsistent sampling frequencies and differences in measured parameters across lakes and time periods. These limitations required additional preprocessing to ensure comparability across datasets and to minimize the impact of incomplete or mismatched data on subsequent analyses.

From an ethical standpoint, all data used within this project were publicly accessible environmental monitoring records. No sensitive information was involved, thus no ethical risks surrounding privacy concerns were associated with the use of this dataset.

Table 2.2.1.*Summary of dataset characteristics used in this study.*

Attribute	Value
Number of Lakes	6
Distance Groups	Near, Mid, Far
Total Observations	3272
Time Span	1972-2025
Number of Variables	25
Data Source	OSMP (Oil Sands Monitoring Program), WQP (Water Quality Portal)
Data Structure	Wide format (one event per row)
Sampling Type	Surface water measurements
Missing Data	Yes
Preprocessing Applied	Standardization, aggregation, median imputation

Table 2.2.2.*Classification of water chemistry variables included in the analysis.*

Category	Variables
Metals	Aluminum Dissolved, Aluminum Total Recoverable, Copper Dissolved, Copper Total Recoverable, Lead Dissolved, Lead Total Recoverable, Mercury Dissolved, Mercury Total, Methyl Mercury, Nickel Dissolved, Nickel Total Recoverable, Vanadium Dissolved, Vanadium Total Recoverable
Nutrients	Nitrogen Kjeldahl Dissolved, Nitrogen Kjeldahl Total, Phosphorus Total, Phosphorus Total Dissolved
Major Ions	Calcium Dissolved, Calcium Total
Physical/Chemical Indicators	pH, pH (Field), Turbidity, Oxygen Dissolved (Field Meter), Oxygen Dissolved % Saturation, Oxygen Biochemical Demand

2.3 Data Cleaning and Pre-processing

To ensure reliability and comparability across datasets, a series of data cleaning and preprocessing steps were applied prior to analysis. The initial cleaning stage focused on addressing obvious data quality issues to reduce noise and improve overall consistency. This included the identification and removal of duplicate rows to prevent bias in subsequent analyses. Columns with a high proportion of missing values were also excluded, as they contained insufficient data to support meaningful statistical analysis (Graham, 2012). In addition, irrelevant metadata columns, such as depth measurements and duplicate identifiers for site names and sampling dates were removed to streamline the dataset and retain only relevant variables.

Further preprocessing steps were undertaken to standardize the dataset and ensure consistency across all variables. A key component of this process involved the standardization of measurement units. While most parameters were reported in micrograms per liter ($\mu\text{g/L}$), some values were recorded in milligrams per liter (mg/L). To ensure comparability, all measurements were converted to a consistent unit ($\mu\text{g/L}$) by applying appropriate conversion factors. In addition to unit standardization, variable names were synchronized across datasets to address inconsistencies in naming. Similar parameters recorded under different column names were aligned to a common format to allow accurate merging and analysis. For example, turbidity measurements reported under different naming conventions were consolidated into a single variable. A summary of the standardized units from the compiled dataset is provided in Table 2.3.1.

Following data cleaning and standardization, additional feature engineering steps were performed to prepare the dataset for analysis. A key addition was the creation of a Distance Group variable, which categorized each lake based on its proximity to oil sands operations (e.g., near-, mid-, and far-field). This categorical feature enabled comparative analysis across spatial groups and supported downstream methods such as clustering and multivariate analysis. To further improve data completeness, additional aggregation steps were applied to reduce missing values arising from inconsistent sampling. Timestamps were simplified by extracting the sampling date and removing time components, allowing observations to be aligned across datasets. Rows corresponding to the same lake and sampling date were then merged into single observations. In cases where multiple measurements of the same parameter existed for a given lake and date, the median value was calculated to represent that sampling event. This approach minimized duplication while preserving a representative value for each parameter.

Following the completion of cleaning, standardization, and feature engineering steps, the final dataset was prepared for analysis. The dataset was organized chronologically based on sampling date to support temporal analysis. Any remaining blank values were left alone as it

showed appropriate. Whereas null columns were disregarded acknowledging the presence as a limitation rather than resulting in potential bias in statistical analysis. The final dataset was structured for consistency across all observations, with each row representing a sampling event and each column corresponding to their chemical or nutrient variable. All preprocessing steps were conducted using python, primarily with the panda's library for data manipulation and transformation.

Table 2.3.1.

Measurement units of water chemistry variables prior to standardization, grouped by unit type.

Units	Variables
µg/L	Aluminum Dissolved, Aluminum Total Recoverable, Copper Dissolved, Copper Total Recoverable, Lead Dissolved, Lead Total Recoverable, Mercury Dissolved, Mercury Total, Methyl Mercury, Nickel Dissolved, Nickel Total Recoverable, Vanadium Dissolved, Vanadium Total Recoverable
mg/L	Calcium Dissolved, Calcium Total, Nitrogen Kjeldahl Dissolved, Nitrogen Kjeldahl Total, Phosphorus Total, Phosphorus Total Dissolved, Oxygen Dissolved (Field Meter), Oxygen Biochemical Demand
Unitless	pH, pH (Field)
NTU	Turbidity
Percentage (%)	Oxygen Dissolved % Saturation